

NAVEEN KARAN R S

Embedded Firmware Engineer | Embedded Systems | ARM Cortex-M

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PROFESSIONAL SUMMARY

Final-year Electronics and Communication Engineering student with 2+ years of hands-on freelance experience designing, developing, and testing embedded firmware applications. Delivered 20+ freelance embedded/IoT projects since 2024 and led firmware development for real-time monitoring and industrial automation systems, including an IoMT ICU monitoring system (96% real-time accuracy) and a priority-based smart ambulance communication framework (40% faster emergency data delivery). Skilled in bare-metal programming, RTOS (FreeRTOS), communication protocols (UART/SPI/I2C/CAN), and hardware-software integration. Seeking an entry-level Embedded/Firmware Engineer role in the semiconductor industry.

TECHNICAL SKILLS

Programming Languages: C, C++, Embedded C, Python

Microcontrollers & Architecture: ARM Cortex-M, STM32, AVR, ESP32, Bare-Metal Programming, Register-Level Programming, Memory-Mapped I/O, DMA, Interrupts, Timers, PWM, ADC, GPIO

Communication Protocols: UART, SPI, I2C, CAN, USB, MODBUS, MQTT

RTOS & Systems: FreeRTOS, QNX Neutrino RTOS, Interrupt Handling (ISR), Task Scheduling, Semaphores, Embedded Linux, Firmware Architecture, Low-Power Design

Development Tools: STM32CubeIDE, QNX Momentics, Keil, PlatformIO, Arduino IDE, Microchip Studio, VS Code, Git, GitHub, MATLAB, Proteus, KiCad

Debugging & Validation: GDB (breakpoints, watchpoints), JTAG, SWD, Oscilloscope, Logic Analyzer, Serial Debugging, Hardware Bring-Up, System-Level Testing

Additional: PCB Design & Layout, Edge ML, WebSockets, QoS Scheduling, Version Control, Technical Documentation

EXPERIENCE

Research and Development Technician | ACM SIGBED – Student Chapter, KARE Apr 2025 – May 2026

- Conducted R&D in Embedded Systems, RTOS, analyzing technical literature and benchmarking hardware architectures.
- Facilitated technical workshops on microcontroller programming, sensor interfacing, and real-time systems for 50+ students.

Embedded Firmware Intern | Kaashiv Info-tech May 2025 – June 2025

- Designed and tested embedded hardware circuits with microcontroller-based control systems, hardware-software integration.
- Conducted validation and system-level integration testing using serial monitor tools to verify communication protocol compliance.

Embedded Systems Developer | HYBIX Group's Startup July 2025 – July 2026

- Led end-to-end development of embedded and IoT solutions, translating requirements into working hardware prototypes.
- Collaborated with clients and cross-functional teams to deliver scalable, production-ready embedded systems.
- Managed firmware development, circuit integration, and testing across multiple microcontroller platforms.

TECHNICAL PROJECTS

IoMT-Based Real-Time Multi-Parameter ICU Patient Monitoring System 2025

Tech Stack: STM32, ESP32, FreeRTOS, MQTT Protocol, UART/I2C Sensor Interfacing, IoMT Sensors, Edge Monitoring

- Developed an IoMT ICU monitoring system tracking with IoMT Sensors achieving 96% real-time monitoring accuracy.
- Developed ESP32 firmware for sensor acquisition, MQTT communication, and health alerts.

DCN-Enabled Priority-Based Smart Ambulance–Hospital Communication Framework 2025

Tech Stack: STM32, ESP32, MQTT Protocol, Embedded C, QoS Scheduling, RFID Priority, Smart Transport

- Designed a priority-based ambulance–hospital communication framework achieving 40% faster emergency data delivery.
- Implemented QoS scheduling with RFID-based traffic prioritization, reducing ambulance intersection delay.

Edge AI-Based Driver Emotion Recognition and Adaptive Vehicle Control System 2026

Tech Stack: STM32, ESP32, Embedded C, CAN Protocol, Python, DeepFace (AI/ML), OpenCV, Motor Driver

- Built a real-time emotion-aware vehicle control system detecting driver state via AI and dynamically adjusting vehicle speed for safety.

Embedded AI Conveyor System for Automated Package Quality Control 2026

Tech Stack: STM32, ESP32, MODBUS Protocol, Embedded C, WebSockets, Edge ML, Industrial Dashboard

- Built a real-time conveyor control system using STM32 with sensor-based automation for object detection and operational safety.

CERTIFICATIONS

- Embedded Systems Internship — Kaashiv Infotech
- Writing Code for ARM® Cortex® Microcontrollers — Microchip University
- 25035 RTOS1: Building Real-Time Applications with FreeRTOS — Microchip University
- Introduction to Embedded Linux — Microchip University
- QNX Neutrino RTOS & Embedded Systems — QNX Pi Square Technologies

ACHIEVEMENTS & LEADERSHIP

- Delivered 20+ freelance projects in embedded systems, IoT, and automation across agriculture, home automation, and industrial monitoring domains since 2024.
- Participated in 15+ hackathons and technical workshops to date, strengthening hands-on skills in embedded systems, IoT, and rapid prototyping.
- Conducted a hands-on workshop on **Next-Gen Embedded Systems and IoT Development**, covering microcontroller programming, sensor interfacing, and real-time applications.

EDUCATION

B.Tech – Electronics and Communication Engineering | Kalasalingam Academy of Research and Education | 2023 – 2027

CGPA: 8.0